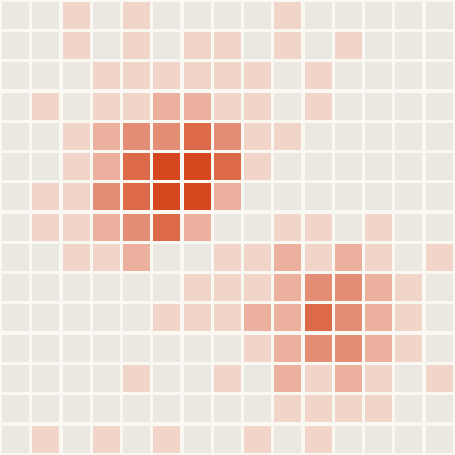
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Cool streets

Three decades of urban heat islands in twelve European cities, seen from orbit



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01

Motivation

01 Motivation

02 Data and method

03 Results

04 Implications

Heat is Europe’s deadliest weather hazard

61 672

heat-related deaths in Europe in the summer of 2022 (Ballester et al. 2023)

+4.6 K

mean summer surface heat island of the twelve cities we study, 2024

1 in 3

Europeans live in a city of more than 100 000 people

Built surfaces store heat by day and release it slowly (Oke 1982); the street, not the weather station, is where people feel it.

02

Data and method

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02 Data and method

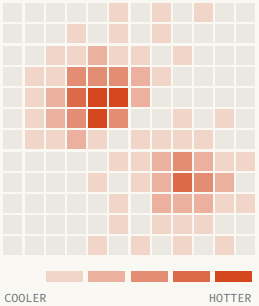
03 Results

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Thirty-four summers of thermal imagery

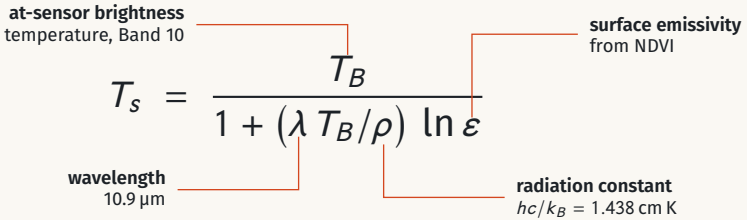
- 1 264 cloud-free summer scenes from Landsat 5, 7, 8 and 9
- Twelve cities, from Lisbon to Warsaw, each with a rural reference ring
- Thermal bands aggregated from 30 m to a 100 m street grid
- 4 812 neighbourhoods linked to canopy, imperviousness and building height

Scenes acquired between June and August, about 10:30 local time.



Land surface temperature, Madrid centre, 3 August 2024

From radiance to surface temperature



Single-channel retrieval after Jiménez-Muñoz and Sobrino (2003). Heat-island intensity is the urban mean minus the rural-ring mean, scene by scene.

03

Results

01 Motivation

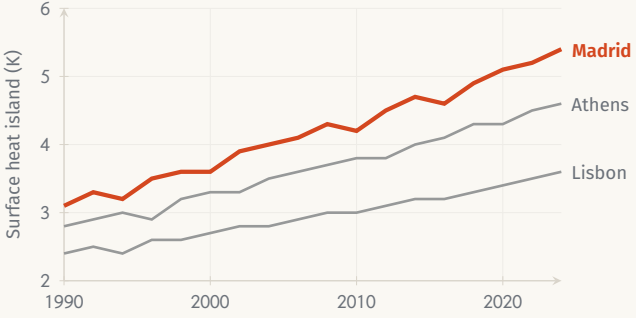
02 Data and method

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RESULTS

Heat islands have grown in every city we studied



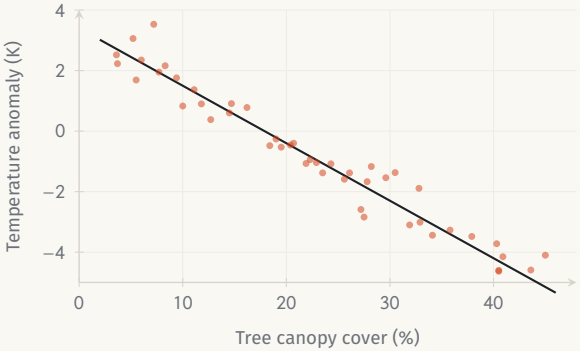
+0.7 K

per decade in Madrid, the fastest of the twelve.

Mean of all summer scenes per two-year window. All twelve trends are positive at $p < 0.01$, in line with the global survey of Peng et al. (2012).

RESULTS

Tree canopy is the strongest cooling lever



−1.9 K

for every ten points of canopy cover.

Each dot is a 100-neighbourhood bin from all twelve cities; the line is the pooled fit. Consistent with Schwaab et al. (2021).

What drives the street-level anomaly

Driver	Effect (K)	95% interval	Variance
Impervious surface, +10 pp	+1.42	[1.31, 1.53]	38%
Tree canopy, +10 pp	−1.87	[−2.01, −1.73]	29%
Roof albedo, +0.1	−0.94	[−1.10, −0.78]	12%
Mean building height, +5 m	+0.36	[0.22, 0.50]	9%
Distance to water, +1 km	+0.21	[0.12, 0.30]	5%

Mixed-effects model with a random intercept per city; $n = 4\,812$ neighbourhoods, $R^2 = 0.71$.

KEY TAKEAWAY

Ten more points of tree canopy cool a street by almost two degrees, more than any other lever a city controls.

Pooled across twelve cities and thirty-four summers of Landsat thermal imagery.

04

Implications

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IMPLICATIONS

Three levers for cooler streets

01

Plant where it is hottest

Target canopy at the 10% of blocks with the highest anomaly; they hold 31% of residents over 65.

02

Lighten the roofs

Raising roof albedo by 0.1 recovers about half the effect of new canopy, at a fraction of the cost.

03

Measure from orbit

The pipeline runs on public Landsat data, so any city can track its heat island every summer.

References

Ballester, Joan et al. (2023). “Heat-related mortality in Europe during the summer of 2022”. In: *Nature Medicine* 29, pp. 1857–1866. DOI: 10.1038/s41591-023-02419-z.

Jiménez-Muñoz, Juan C. and José A. Sobrino (2003). “A generalized single-channel method for retrieving land surface temperature from remote sensing data”. In: *Journal of Geophysical Research: Atmospheres* 108.D22. DOI: 10.1029/2003JD003480.

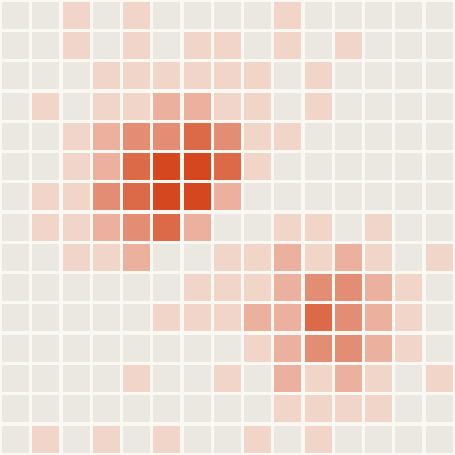
Oke, Timothy R. (1982). “The energetic basis of the urban heat island”. In: *Quarterly Journal of the Royal Meteorological Society* 108.455, pp. 1–24. DOI: 10.1002/qj.49710845502.

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QUESTIONS

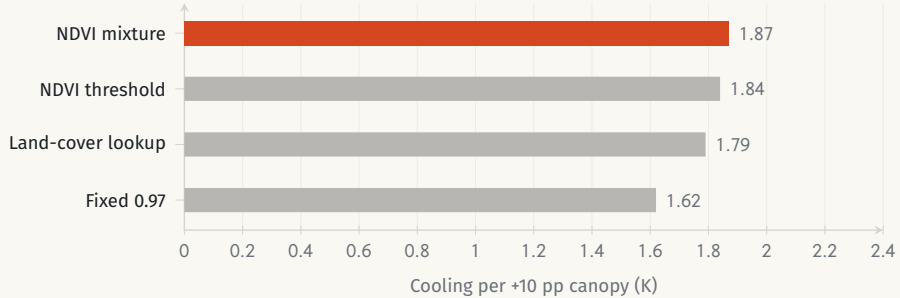
Thank you



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data and code github.com/northfield-climate/cool-streets

Backup: robust to the emissivity scheme



The main analysis uses the NDVI mixture scheme (highlighted). Every alternative keeps canopy as the strongest cooling lever.